

React Day 3 - Forms and useState

Day 3 Overview

Topics Covered

- ✓ Creating Forms in React
 - ✓ Form Styling
 - ✓ Controlled Components
 - ✓ useState Hook
 - ✓ Handling Input Fields
 - ✓ Handling Select Dropdown
 - ✓ onChange Event
 - ✓ Form Submission
 - ✓ preventDefault()
 - ✓ Creating Objects from Form Data
-

Folder Structure

```
None
src
|
├── components
|   ├── Addnewdoctor.jsx
|   └── styles.css
|
└── App.jsx
```

└─ main.jsx

Part 1: Creating Doctor Form UI

Technical Definition

A **Form** is a collection of input fields used to collect user data.

Addnewdoctor.jsx

None

```
function Addnewdoctor() {  
  return (  
    <div>  
      <h1>Add New Doctor</h1>  
  
      <form className="form-container">  
        <input  
          type="text"  
          placeholder="Enter Doctor Name"  
          className="text-field"  
        />  
  
        <input  
          type="text"  
          placeholder="Enter Age"  
          className="text-field"  
        />  
      </form>  
    </div>  
  );  
}
```

```
    <select>
      <option>Select Gender</option>
      <option value="Male">Male</option>
      <option value="Female">Female</option>
    </select>

    <input
      type="text"
      placeholder="Enter Specialization"
      className="text-field"
    />

    <input
      type="text"
      placeholder="Enter Salary"
      className="text-field"
    />

    <button>Add Doctor</button>
  </form>
</div>
);
}

export default Addnewdoctor;
```

Styling

styles.css

CSS

```
.form-container {  
  width: 33%;  
  margin: auto;  
  border: 2px solid black;  
  padding: 20px;  
}  
  
.text-field,  
select,  
.form-container button {  
  display: block;  
  padding: 10px;  
  width: 100%;  
  margin: 10px;  
}  
  
.form-container button {  
  background: black;  
  color: white;  
}
```

Part 2: Using useState for Form Inputs

Technical Definition of useState

`useState` is a React Hook used to create and manage state inside functional components.

Syntax

None

```
const [state, setState] = useState(initialValue);
```

Creating State Variables

None

```
const [name, setName] = useState("");  
const [gender, setGender] = useState("");  
const [age, setAge] = useState("");  
const [salary, setSalary] = useState("");  
const [specialization, setSpecialization] = useState("");
```

Technical Definition of Controlled Component

A **Controlled Component** is a form element whose value is controlled by React State.

Controlled Inputs

None

```
<input  
  value={name}  
  onChange={(e) => setName(e.target.value)}  
  type="text"  
>
```

None

```
<input
  value={age}
  onChange={(e) => setAge(e.target.value)}
  type="text"
/>
```

None

```
<select
  value={gender}
  onChange={(e) => setGender(e.target.value)}
>
  <option>Select Gender</option>
  <option value="Male">Male</option>
  <option value="Female">Female</option>
</select>
```

None

```
<input
  value={specialization}
  onChange={(e) => setSpecialization(e.target.value)}
  type="text"
/>
```

None

```
<input
```

```
value={salary}
onChange={(e) => setSalary(e.target.value)}
type="text"
/>
```

Part 3: Form Submission

Technical Definition of onSubmit

`onSubmit` is an event that executes when a form is submitted.

Technical Definition of preventDefault()

`preventDefault()` prevents the browser's default form submission behavior and page refresh.

Submit Function

```
None
function handleSubmit(e) {
  e.preventDefault();

  let newdoctor = {
    name,
    age,
    gender,
    salary,
    specialization,
```

```
};

console.log(newdoctor);
}
```

Connecting Form with Submit Function

```
None
<form
  className="form-container"
  onSubmit={handlesubmit}
>
```

Submit Button

```
None
<button type="submit">
  Add Doctor
</button>
```

OUTPUT

Add New Doctor

Select Gender 

Add Doctor

Day 3 Summary

Concepts Learned

- Forms in React
- Form Styling
- useState Hook
- State Variables
- Controlled Components
- onChange Event
- Handling Input Fields
- Handling Dropdowns
- onSubmit Event
- preventDefault()
- Creating Objects
- Form Data Collection